



Anxiolytic and Hypnotic drugs

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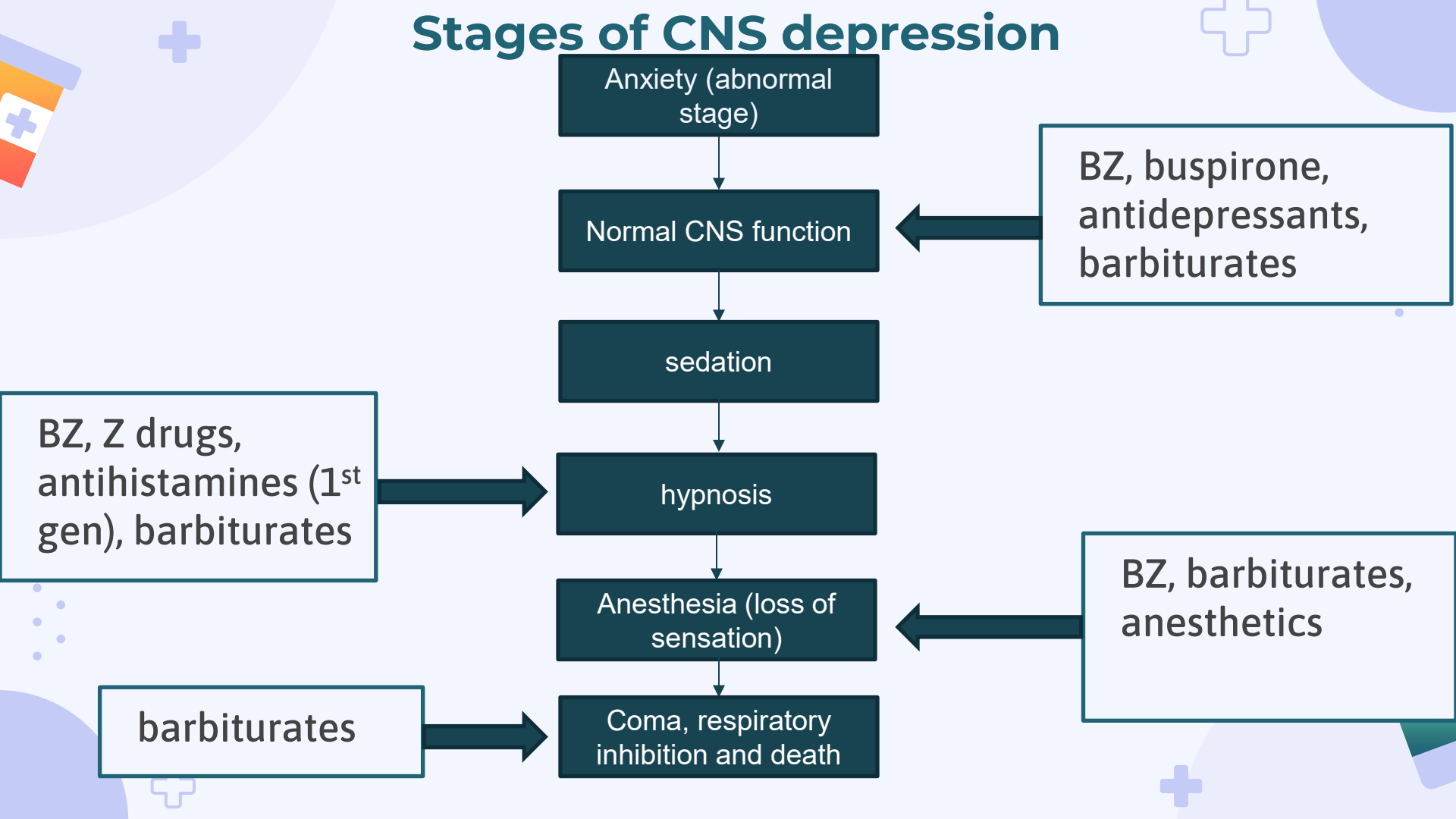
01 Overview



Introduction

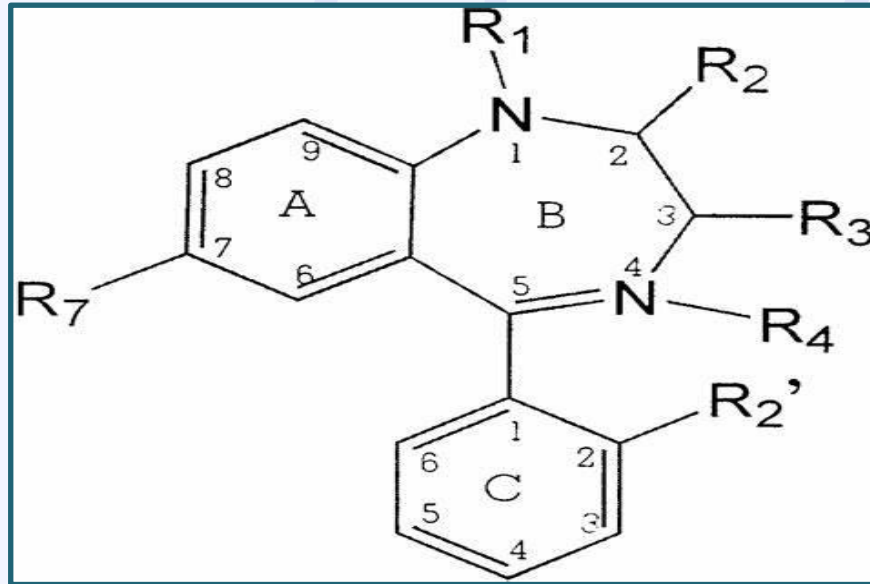
- Disorders involving anxiety are among the most common mental disorders. Anxiety is an unpleasant state of tension, apprehension, or uneasiness (a fear that arises from either a known or an unknown source).
- The physical symptoms of severe anxiety are similar to those of fear (such as tachycardia, sweating, trembling, and palpitations) and involve sympathetic activation. Episodes of mild anxiety are common life experiences and do not warrant treatment.
- severe, chronic, debilitating anxiety (most days for at least 6 months) may be treated with antianxiety drugs (sometimes called anxiolytics) and/or some form of psychotherapy.

Stages of CNS depression



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Benzodiazepines



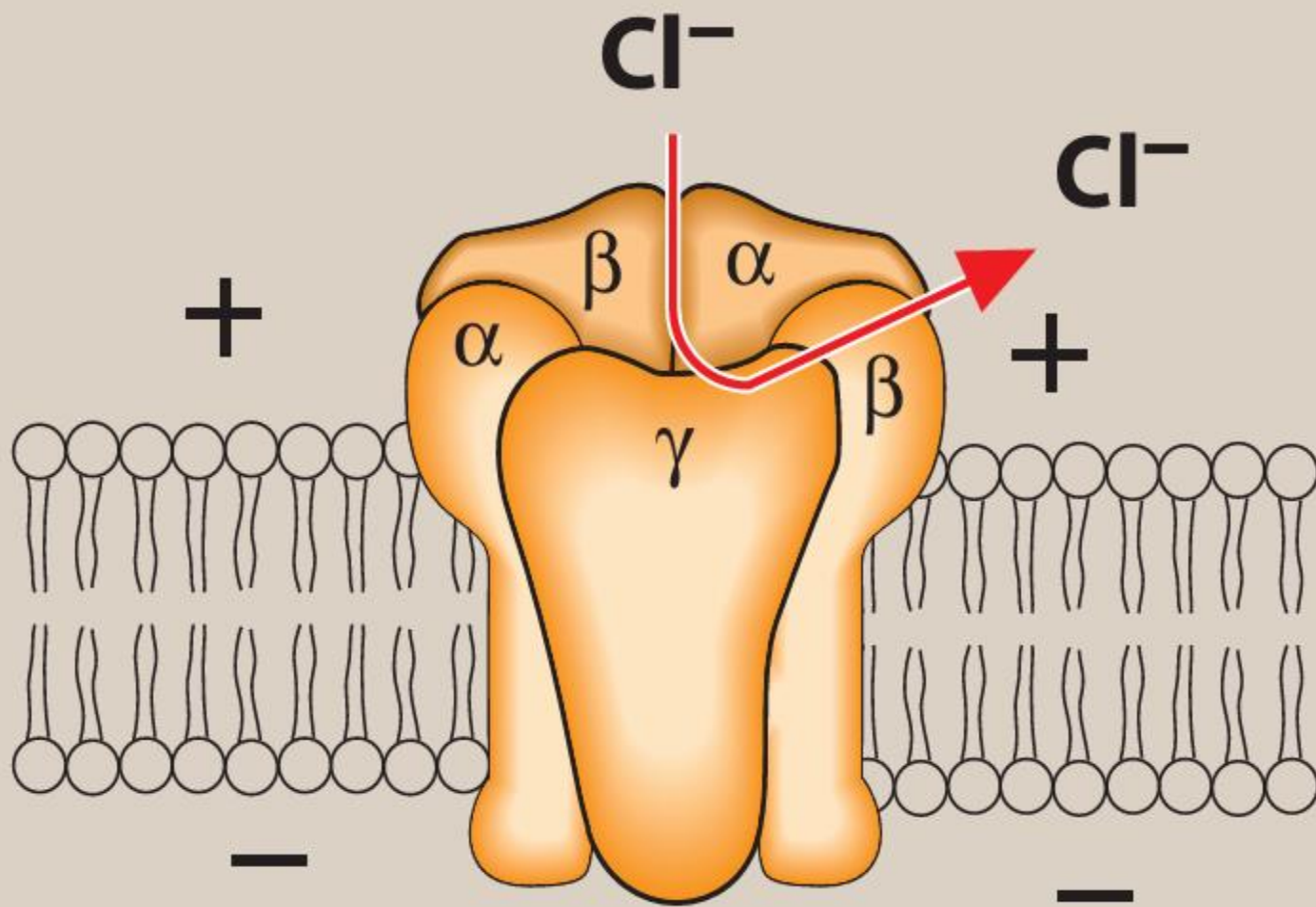
Introduction

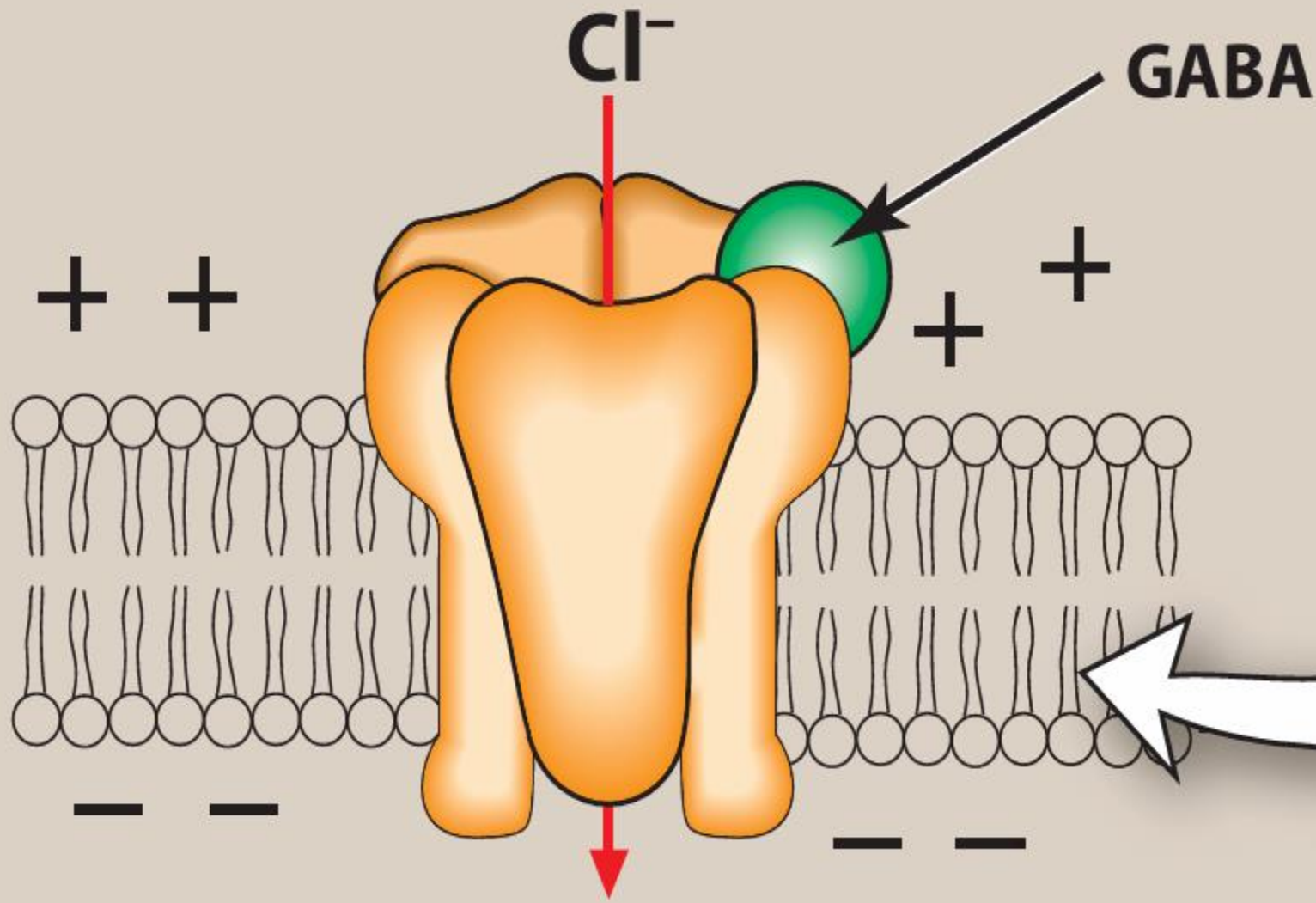
- Benzodiazepines are widely used anxiolytic drugs. They have largely replaced barbiturates in the treatment of anxiety and insomnia, because benzodiazepines are generally considered to be safer and more effective.
- However, they are not the 1st choice in treatment of chronic anxiety or chronic insomnia, they are used for short term management or as adjuvant to standard treatment. This limitation is due to risk of abuse and dependence often associated with benzodiazepines.

Mechanism of action

- GABA is the major inhibitory neurotransmitter in the central nervous system (CNS). The GABA receptors are composed of a combination of five α , β , and γ subunits that span the postsynaptic membrane
- Binding of GABA to its receptor triggers an opening of the central ion channel, allowing chloride through the pore. The influx of chloride ions causes hyperpolarization of the neuron and decreases neurotransmission by inhibiting the formation of action potentials.

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- Benzodiazepines modulate GABA effects by binding to a specific, high-affinity site (distinct from the GABA-binding site) located at the interface of the α subunit and the γ subunit on the GABA receptor.
 - Note: These binding sites are sometimes labeled “benzodiazepine (BZ) receptors.”
 - Benzodiazepines increase the frequency of channel openings produced by GABA.
 - This mechanism is dubbed allosteric modulation, since benzodiazepines do not affect the opening of the ion channel but cause configurational changes in GABA receptor which enhances GABA binding.
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Benzodiazepine

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GABA

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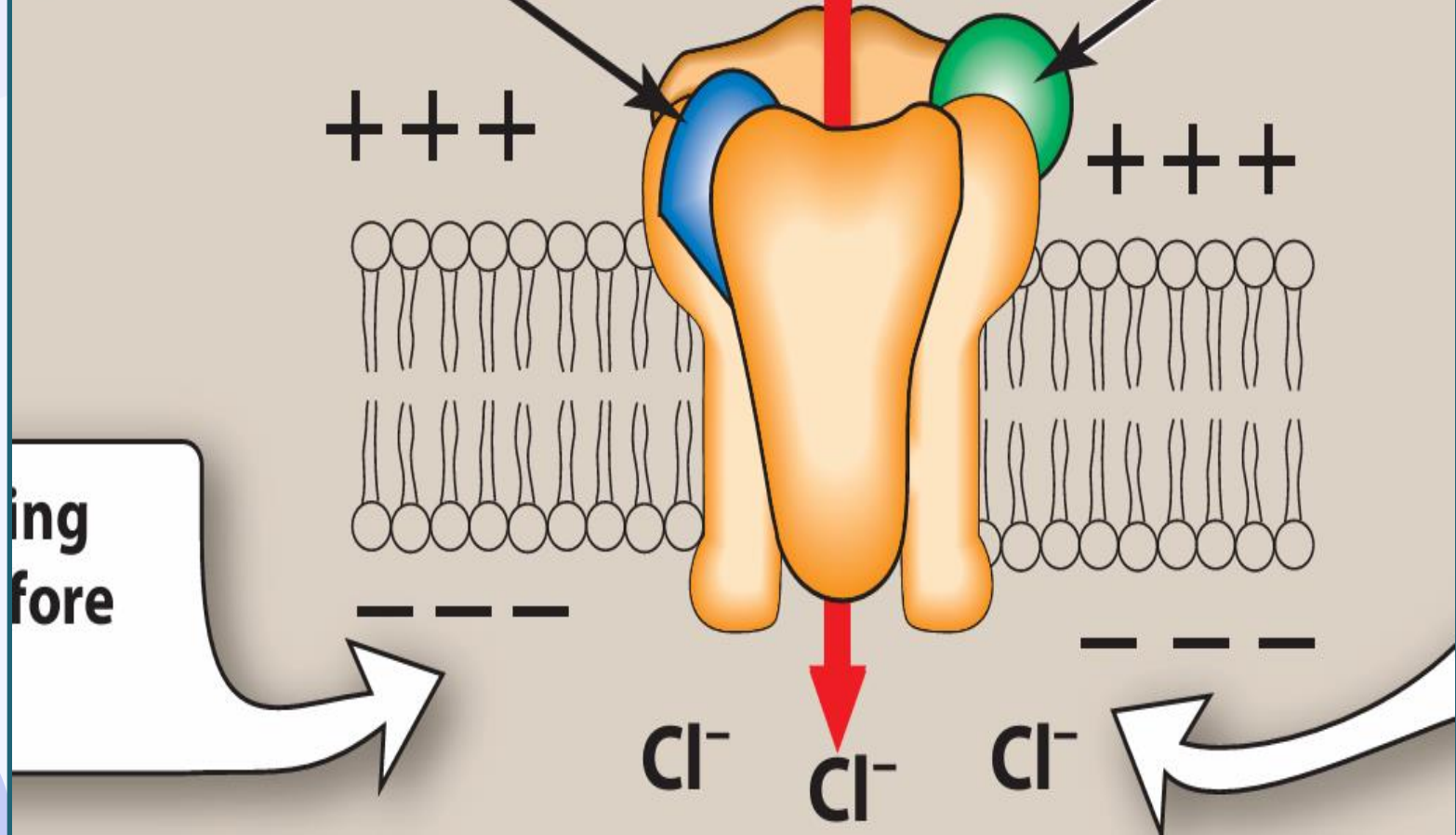
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Types of BZ

- The individual benzodiazepines show small differences in their relative anxiolytic, anticonvulsant, and sedative properties. However, the duration of action varies widely among this group, and pharmacokinetic considerations are often important in choosing one benzodiazepine over another.
- Short acting BZ ($t_{1/2}$ 3-8 hours)
 - Alprazolam (tab)
 - Midazolam (inj)
 - Triazolam
- intermediate acting BZ ($t_{1/2}$ 10-20 hour)
 - Diazepam (tab) (supp) (inj)
 - Clonazepam
 - Oxazepam (a metabolite of Diazepam)
- Long acting BZ ($t_{1/2}$ 1-3 days): chlordiazepoxide

Pharmacokinetics

- Absorption and distribution:

- The benzodiazepines are lipophilic. They are rapidly and completely absorbed after oral administration, distribute throughout the body and penetrate into the CNS.

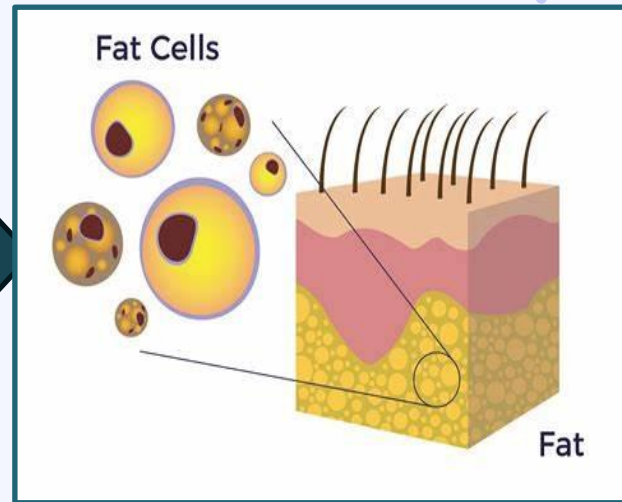
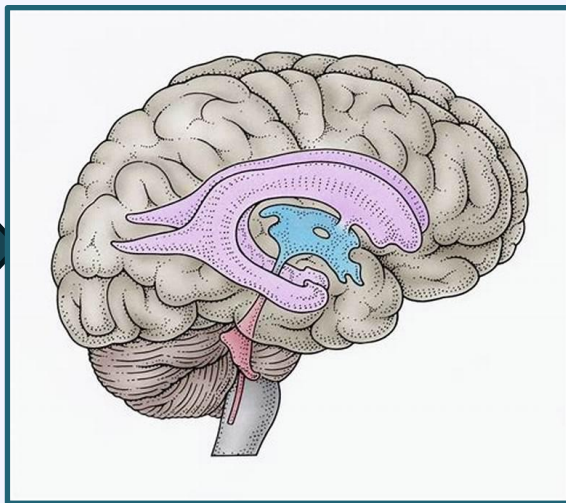
- Duration of action:

- The half-lives of the benzodiazepines are important clinically, because the duration of action may determine the therapeutic usefulness
- However, the clinical duration of action does not correlate with the actual half-life (otherwise, a dose of diazepam could conceivably be given only every other day, given its active metabolites). This may be due to receptor dissociation rates in the CNS and subsequent redistribution to fatty tissues and other areas.

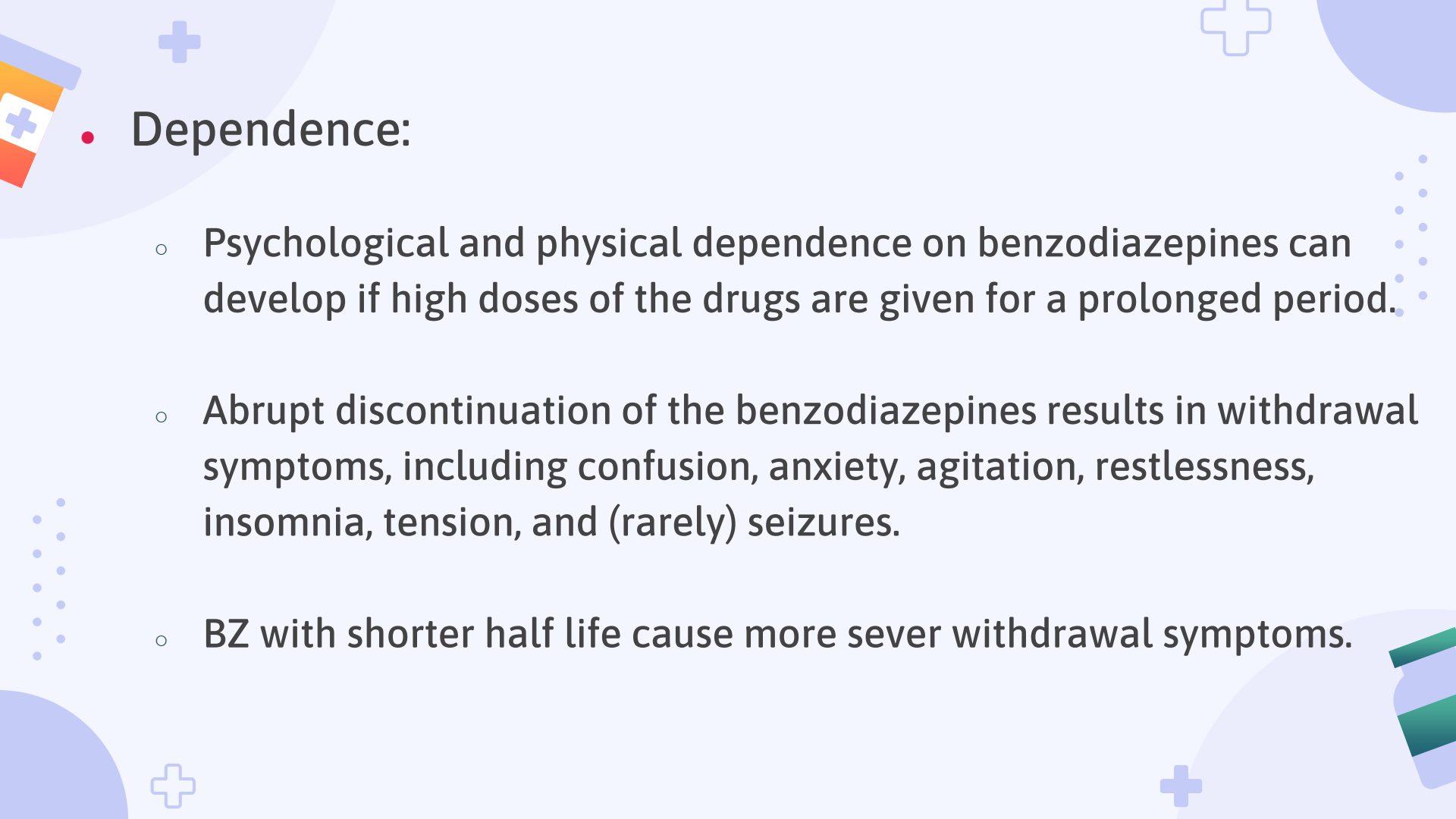


• Metabolism and excretion:

- Most benzodiazepines are metabolized by the hepatic microsomal system to compounds that are also active.
 - For these benzodiazepines, the apparent half-life of the drug represents the combined actions of the parent drug and its metabolites.
 - Termination of action is through distribution and further metabolism by conjugation into glucuronide conjugates to make them more hydrophilic.
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- Dependence:

- Psychological and physical dependence on benzodiazepines can develop if high doses of the drugs are given for a prolonged period.
 - Abrupt discontinuation of the benzodiazepines results in withdrawal symptoms, including confusion, anxiety, agitation, restlessness, insomnia, tension, and (rarely) seizures.
 - BZ with shorter half life cause more severe withdrawal symptoms.
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- Pregnancy and lactation:

- BZ cross the placenta and are teratogenic, they also are secreted in breast milk and thus a nursing mother can expose the infant to BZ.

Therapeutic uses

- Anxiety disorders (low dose):
 - Benzodiazepines are effective for the treatment of the anxiety symptoms secondary to panic disorder, generalized anxiety disorder (GAD). obsessive-compulsive disorder, and extreme anxiety associated with phobias, such as fear of flying.
 - The benzodiazepines are also useful in treating anxiety related to depression and schizophrenia.



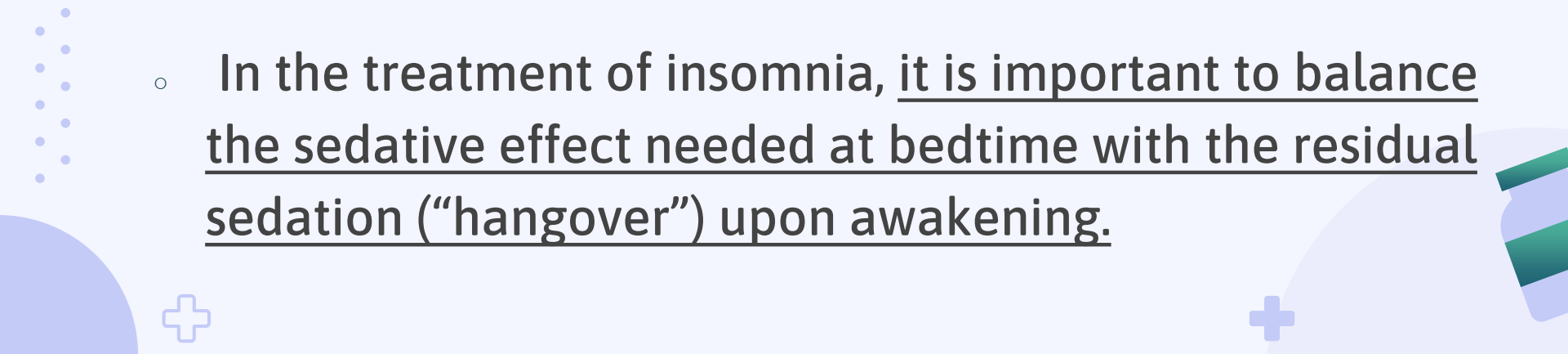
• Anxiety disorders:

- The longer-acting agents are often preferred in those patients with anxiety that may require prolonged treatment. The antianxiety effects of the benzodiazepines are less subject to tolerance than the sedative and hypnotic effects.
- For panic disorders, alprazolam is effective for short and long-term treatment, although it may cause withdrawal reactions in about 30% of patients.





• Sleep disorders (high dose):

- These agents decrease the latency to sleep onset and increase stage II of non-rapid eye movement (REM) sleep. Both REM sleep and slow-wave sleep are decreased.
 - In the treatment of insomnia, it is important to balance the sedative effect needed at bedtime with the residual sedation (“hangover”) upon awakening.
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Sleep disorders:

- Short acting agents with rapid onset of action are effective in treating individuals who have problems falling asleep but carry the risk of withdrawal and rebound insomnia after discontinuation.
 - Long and intermediate acting agents are useful for patients who experience frequent awakenings or have difficulty staying asleep, however, drug effect can extend to day time which will cause day time drowsiness (hang-over effect)
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Amnesia:

- Anterograde amnesia: Temporary impairment of memory with use of the benzodiazepines. In which The ability to learn and form new memories is also impaired.
 - The shorter-acting agents are often employed as premedication for anxiety-provoking and unpleasant procedures (endoscopy) which makes the patient more compliant and has little to no memory of the experience.
 - Midazolam is the most commonly used BZ to provide  this level of sedation. 
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Seizures:

- The use of BZ in treatment of epilepsy is no longer accepted due to development of better and much safer anti-epileptic drugs.
 - Diazepam is the drug of choice in terminating status epilepticus.
 - Diazepam supp are used to treat seizures associated with febrile seizures.
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• Muscular disorders:

- Diazepam is useful in the treatment of skeletal muscle spasms, such as occur in muscle strain, and in treating spasticity from degenerative disorders, such as multiple sclerosis and cerebral palsy.
 - However, the experience with BZ use in muscular disorders is limited.
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Adverse effects

- Physical dependence and tolerance
- Rebound insomnia after discontinuation.
- Hangover effect: a state of psychomotor disturbance that occurs the following day after use of long acting drugs (I.E residual effect)
- Anterograde amnesia and memory disturbance.
- CYP inhibitors can prolong the action of these drugs.
- Benzodiazepines are very safe as toxic dose is 1000 fold the therapeutic dose.

BZ antagonist

• Flumazenil

- is a GABA receptor antagonist that can rapidly reverse the effects of benzodiazepines. The drug is available for intravenous (IV) administration only.
- Administration of flumazenil may precipitate withdrawal in dependent patients or cause seizures if a benzodiazepine is used to control seizure activity.

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03 Other anxiolytic drugs



Anti-depressants

- SSRI and SNRI are considered the first line treatment for chronic anxiety, GAD, and OCD due to absence of risk of dependence and tolerance.
- Escitalopram, and venlafaxine are approved for these conditions which may be used alone or in combination with a benzodiazepine.
- This combination is beneficial for sever cases as SSRI and SNRI require at least 4-6 weeks to produce a clinically significant effect.

Buspirone

- Buspirone is useful for the chronic treatment of GAD and has an efficacy comparable to that of the benzodiazepines. It has a slow onset of action and is not effective for short-term or “as-needed” treatment of acute anxiety states.
- The actions of buspirone appear to be mediated by serotonin (5-HT_{1A}) and (5-HT_{2A}) receptors
- buspirone lacks the anticonvulsant and muscle-relaxant properties of the benzodiazepines.
- The frequency of adverse effects is low, with the most common effects being headaches, dizziness, nervousness, nausea, and light-headedness. Sedation and psychomotor and cognitive dysfunction are minimal, and dependence is unlikely.

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Clinical application: Management of GAD

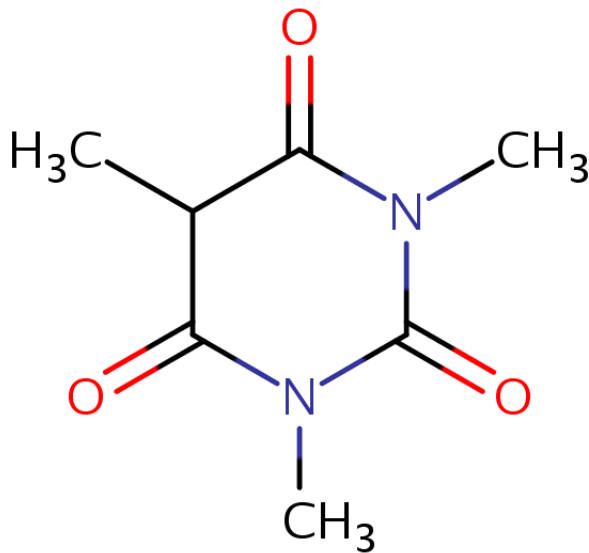
- Psychotherapy is the 1st step in treatment of GAD and OCD (acute or chronic).
- Acute anxiety is relieved by the use of BZ (preferably a short acting BZ like alprazolam for a period of 1-2 week)
- Chronic anxiety:
 - SSRI or SNRI, clinical effect is observed within 4-6 weeks.
 - If response is not adequate:
 - Switch to another SSRI or SNRI
 - Add BZ or buspirone or antipsychotic agent



Clinical application: Management of GAD

- The combination is used for severe GAD with anxiety symptom
- Week 1: SSRI (Escitalopram) + intermediate acting BZ (lorazepam).
- Week 4-6: SSRI (Escitalopram) + tapering (dose reduction of) lorazepam.
- Short acting BZ are not preferred in chronic treatment due to severe withdrawal symptoms and difficulty in drug tapering.

03 Barbitutes



Introduction

- The barbiturates were formerly the mainstay of treatment to sedate patients or to induce and maintain sleep.
- Today, they have been largely replaced by the benzodiazepines, primarily because:
 - Barbiturates induce rapid tolerance and physical dependence.
 - They are associated with very severe withdrawal symptoms.
 - Lethal in overdose.

Mechanism of action

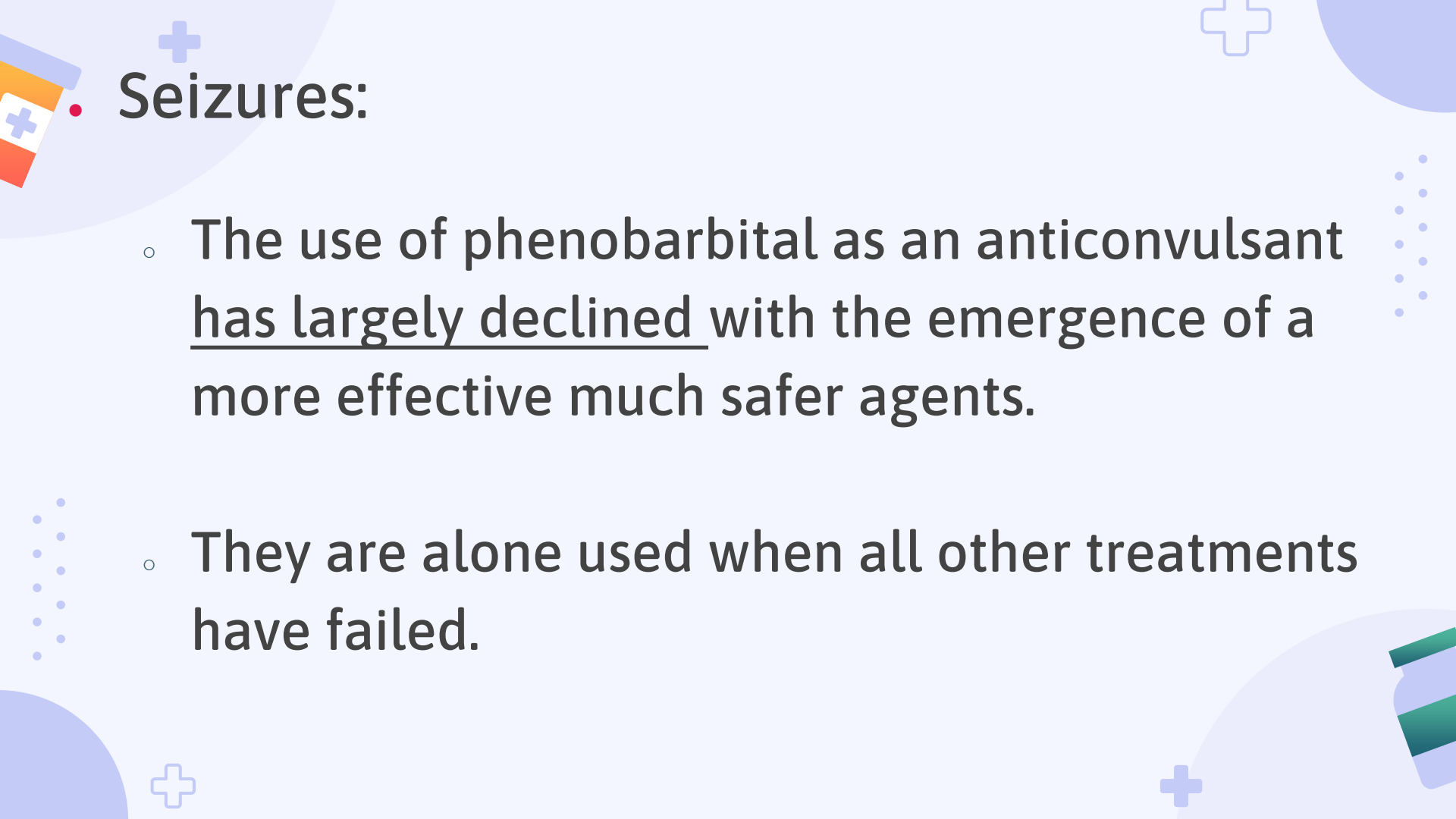
- Allosteric modulation of GABA receptor by binding to a site that is distinct from both GABA site and BZ site. This produces an action similar to BZ action.
- They also block the release of the excitatory glutamate causing further CNS inhibition and decreased neuronal activity.
- At low doses barbiturates cause sedation (calming, relaxing effect), at high doses they cause hypnosis, anesthesia.
- Overdoses can cause coma and death with inhibition of the respiration process.

Types of Barbiturates

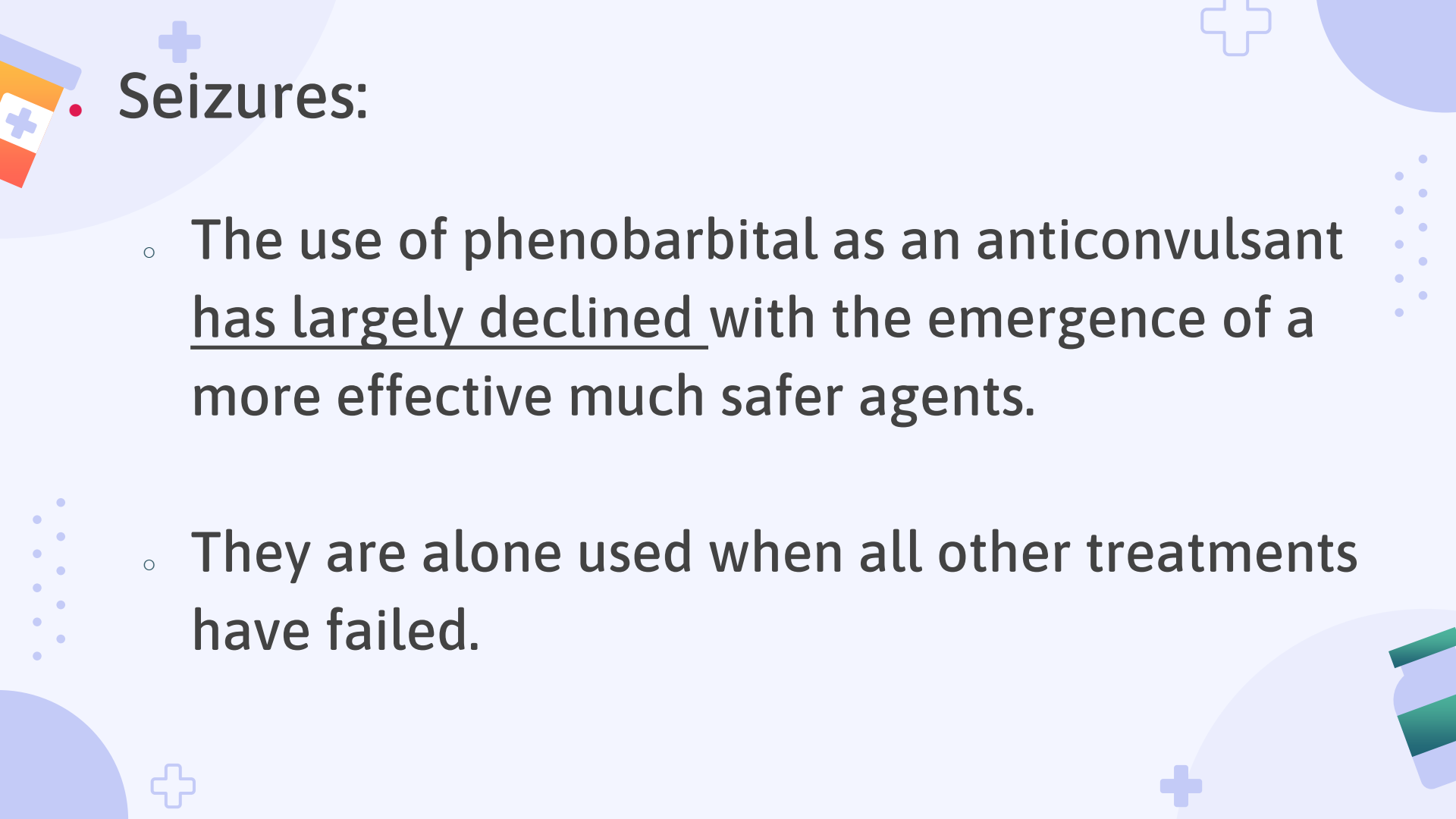
- Ultrashort acting Barbiturates (onset is within a few seconds)
 - Thiopental
- Short acting Barbiturates ($t_{1/2}$ 3-8 hours)
 - Pentobarbital
 - Secobarbital
- Long acting Barbiturates ($t_{1/2}$ 1-2 days):
 - Phenobarbital

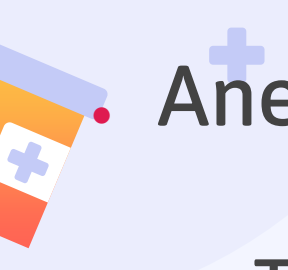
Therapeutic uses

- Anxiety disorders, sedative/hypnotic:
 - the use of barbiturates for insomnia is no longer generally accepted, given their adverse effects and potential for tolerance.



Seizures:

- The use of phenobarbital as an anticonvulsant has largely declined with the emergence of a more effective much safer agents.
 - They are alone used when all other treatments have failed.
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Anesthesia:

- Thiopental was used to induce anesthesia, however, it has been replaced by better and safer agents.
 - I.V pentobarbital is used to induce a “barbiturate coma” in patients with traumatic brain injury or those awaiting skin grafts in severe burns.
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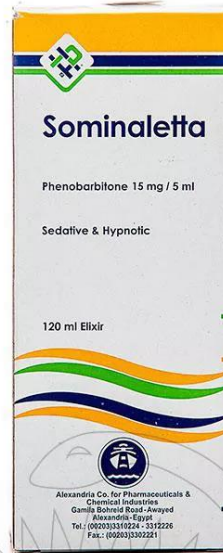


Adverse effects and pharmacokinetics



- The kinetics and adverse effects of barbiturates and BZ are very similar, the difference is that:
 - Barbiturates are associated with very severe withdrawal effects, and can lower the seizure threshold.
 - Barbiturates are CYP-450 inducers and can reduce the activity of many drugs.
 - Because they are CYP-450 inducers they can be used to breakdown bilirubin in neonates with hyperbilirubinemia, however, U.V light therapy is the 1st choice in this condition.
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04 Other hypnotic drugs



Z drugs

- Zolpidem, zaleplon, and eszopiclone are collectively called the Z drugs.
- Non of these drugs is structurally related to BZ, but They do not significantly alter the various sleep stages and, hence, are often the preferred hypnotics. This may be due to their relative selectivity for the BZ receptor.
- they exhibit no anticonvulsant or muscle-relaxing properties or anxiolytic effect.
- They show few withdrawal effects, exhibits minimal rebound insomnia, and little tolerance occurs with prolonged use.

Zolpidem

- Zolpidem is rapidly absorbed from the gastrointestinal (GI) tract, and it has a rapid onset of action and short elimination half-life (about 2 to 3 hours). It provides a hypnotic effect for approximately 5 hours.
- Zolpidem undergoes hepatic oxidation by the CYP450 system to inactive products , and so it is subject to CYP450 related adverse effects.
- Adverse effects of zolpidem include:
 - Nightmares
 - Agitation
 - Daytime drowsiness

Zaleplon

- Zaleplon is an oral nonbenzodiazepine hypnotic similar to zolpidem;
- zaleplon causes fewer residual effects on psychomotor and cognitive function compared to zolpidem or the benzodiazepines.
- This may be due to its rapid elimination, with a half-life of approximately 1 hour. The drug is metabolized by CYP3A4.

Eszopiclone

- Eszopiclone is an oral nonbenzodiazepine hypnotic that also acts on the BZ1 receptor.
- It has been shown to be effective for insomnia for up to 6 months.
- Eszopiclone is rapidly absorbed (time to peak, 1 hour), extensively metabolized the CYP450 system

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Antihistamines

- Some antihistamines with sedating properties such as:
 - Diphenhydramine
 - Hydroxyzine
 - Doxylamine
- they are effective in treating mild types of situational insomnia.

Antidepressants

- The use of sedating antidepressants with strong antihistamine profiles has been ongoing for decades.
- Doxepin an older tricyclic agent with SNRI mechanisms of antidepressant and anxiolytic action, was recently approved at low doses for the management of insomnia.

Clinical application: Management insomnia

- Short term insomnia (less than 3 months) can be caused by an acute life stressor (such as a death in the family) and chronic insomnia both benefit from employing sleep hygiene and stress management.
- Short term insomnia can be managed by: BZ, Z drugs, and antihistamines.
- Long term and chronic insomnia will require psychic and medical evaluation. Certain medical conditions likes: sleep apnea, pulmonary disease, HF, chronic pain can be the cause of insomnia.
- The use of Z drugs, and anti-depressants are employed in chronic insomnia as they carry a lower risk of dependence and tolerance than BZ, however, the risk with Z drugs is still present since they are controlled substance.

Cases

- An 18-year-old male is having difficulty sleeping because of the death of his grandfather. He is given a benzodiazepine that does which of the following?
 - A. Binds to serotonin 5-HT 1 receptors
 - B. Binds to GABA A receptors
 - C. Is an antagonist at α -adrenoceptors
 - D. Is an antagonist at dopamine D 2 receptors

Answers

- B. Benzodiazepines enhance the binding of GABAA to its receptor, which increases the permeability of chloride.
- The benzodiazepines do not relieve pain but may reduce the anxiety associated with pain.

Cases

- Which one of the following is a short-acting hypnotic?
 - A. Phenobarbital.
 - B. Diazepam.
 - C. Chlordiazepoxide.
 - D. Triazolam.

Answers

- D. Triazolam is a short-acting drug. It has little daytime sedation. The other drugs listed are longer acting

Cases

- A 45-year-old man who has been injured in a car accident is brought into the emergency room. His blood alcohol level on admission is 275 mg/dL. Hospital records show a prior hospitalization for alcohol-related seizures.
- His wife confirms that he has been drinking heavily for 3 weeks. What treatment should be provided to the patient if he goes into withdrawal?
- A. None. B. Lorazepam. C. Pentobarbital. D. Phenytoin. E. Buspirone.

Answers

- B. It is important to treat the seizures associated with alcohol withdrawal. Benzodiazepines, such as chlordiazepoxide, diazepam, or the shorter-acting lorazepam, are effective in controlling this problem.
- They are less sedating than pentobarbital or phenytoin